Robust LOS/NLOS Identification for UWB Signals Using Improved Fuzzy Decision Tree Under Volatile Indoor Conditions

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***Abstract*— Ultra-wideband (UWB) is a very promising indoor wireless positioning technology. However, in the harsh and volatile indoor environment, the propagation of UWB signals is vulnerable to non-line-of-sight (NLOS) conditions, and the contaminated range measurements will degrade the accuracy for UWB localization. Therefore, it is necessary to identify line-of-sight (LOS)/NLOS. Recent studies mainly focus on the identification of UWB signal propagation conditions by using channel impulse response (CIR) or extracted channel statistical features. However, these studies usually only focus on specific indoor environment or stable indoor conditions. In fact, the indoor environment is harsh and changeable. In order to deal with the dynamic and uncertain information of the indoor environments, this article proposes a robust method to identify LOS/NLOS using fuzzy decision tree (FDT) based on the Bayesian optimization. The proposed method first extracts the classification features from the CIRs and then fuzzifies the features. Finally, it combines Bayesian optimization to construct the FDT, so as to identify the propagation conditions of UWB signals. The experimental results show that the identification accuracy of the proposed method is higher than 90% in both static and dynamic experiments, and the overall performance is excellent. Compared with other methods, it has certain advantages and robustness.**

***Index Terms*— Bayesian optimization, fuzzy decision tree (FDT), non-line-of-sight (NLOS), robust, ultra-wideband (UWB).**

I. Introduction

With the rapid development of the Internet-of-Things (IoT) technology, demand for location-based services (LBSs) is also increasing. Source localization is vital to enable LBS. Due to the occlusion of signals by building roofs and walls, the Global Navigation Satellite System (GNSS) can no longer provide accurate and reliable position information indoors [1]. Therefore, in this case, accurate indoor positioning has become one of the research hot spots of LBS. Currently, indoor wireless positioning technologies mainly include radio frequency identification (RFID), Bluetooth low energy (BLE), Wi-Fi, ultrasonic, and ultra-wideband (UWB) [2], [3], [4], [5], [6]. In many indoor positioning technologies, UWB positioning technology has the advantages of low transmission power and thus low power consumption and low interference to other systems, high time resolution and thus insensitive to multipath interference, and strong penetration ability, so it has become a very promising high-accuracy indoor positioning technology.

The non-line-of-sight (NLOS) propagation is the main error source of UWB indoor positioning. In the presence of obstacles (such as wooden doors, pedestrians, and thin and thick walls) along the signal propagation path, due to the occlusion of obstacles, signal refraction, reflection, and penetration occur. In these cases, a positive bias will be produced in the range measurement, which will lead to a significant decline in positioning accuracy. Therefore, in order to ensure high-accuracy positioning, it is necessary to identify NLOS measurements and mitigate their adverse effects. Researchers have done a lot of work on the problem of NLOS identification for UWB signals in indoor positioning. Similar to [7], NLOS identification methods can be divided into three types: range-based methods [8], [9], [10], location-based methods [11], and channel-based methods [12]. Range-based methods mainly use the variance of range measurements and the probability density function (PDF) to distinguish NLOS from line-of-sight (LOS). Most of these methods require some prior knowledge and rely on a large number of range measurements. Therefore, range-based methods are usually not suited for real-time positioning. Location-based methods mainly use external auxiliary information, such as map, geometric location information. Although these methods do not require prior knowledge, the auxiliary information needs to be updated frequently when the indoor environment changes.

Channel-based methods use the statistical features of channel impulse response (CIR) or received UWB signal waveforms, such as root mean square, kurtosis (KUR), and maximum amplitude (MA), to distinguish LOS from NLOS. Based on the extracted features, the NLOS identification can be completed by a preset reasonable threshold [12], [13]. However, when outliers appear in the samples, this strategy can make mistakes, and the threshold usually depends on actual environment. Therefore, the method is simple but has limited identification accuracy. In [14], a fuzzy comprehensive

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evaluation (FCE) method is proposed, which collects the measurements of UWB signals offline under different channel propagation conditions and then calculates the similarity with the online measurements to identify NLOS. Although the FCE method can effectively reduce the computational complexity, it needs to collect enough offline data. In addition, with the development of machine learning (ML), its application has been extended to NLOS identification for UWB signals. The supervised classification method of traditional ML constructs the mapping relationship between channel statistical characteristics and signal propagation conditions through the training set [15], [16], [17], [18], [19], [20]. Therefore, offline data, training, and selection of model are crucial. However, collecting and labeling offline data are time-consuming and laborious, and offline data need to be updated frequently to adapt to the dynamic indoor environments. To solve this problem, semi-supervised classification and unsupervised classification without labeling manually are proposed for NLOS identification [21], [22], [23]. However, in contrast, the accuracy of unsupervised classification for NLOS identification is limited. In recent years, with the development of deep learning (DL), researchers have also done some related work. In view of the advantages of convolutional neural network (CNN) in extracting image features, some studies focus on the time–frequency analysis of the received UWB signal by using methods and theories in the field of signal processing. The converted time–frequency images or color coefficient images are used to train the CNN and then identify NLOS conditions [24], [25]. In addition, CNN and long short-term memory (LSTM)-based methods for NLOS identification have been proposed. The features extracted by CNN from CIRs are used as the input of LSTM. The experimental results show that the classification accuracy is further improved [26], [27], [28].

ML technology has its unique advantages, but this kind of method relies too much on offline data, and the dataset supporting model training needs to be sufficient. It is time-consuming and laborious to update offline data frequently and retrain the model in harsh and volatile indoor environments. In practical applications, it is one of the research hot spots to effectively improve the generalization ability of NLOS identification algorithms while ensuring accuracy requirements. In harsh indoor environment, uncertainties are added to the measurement data due to factors, such as random noise and human activity interference. Additionally, due to the complex propagation channel, the channel types represented by offline data may not fully reflect the real situation. It is difficult for ML methods to obtain good performance in this case. For uncertainty, there are two explanations. One is the inherent noise in data, that is, the inevitable error in the measurement; second, the prediction ability of the model itself to unknown input data is limited because of poor training or insufficient and inaccurate training data. Unfortunately, the existing research rarely fully considers the uncertainty in practical applications. However, this uncertainty can be considered by applying a fuzzy classifier [29]. In addition, some ML methods have weak interpretability in LOS/NLOS identification. To solve these problems, fuzzy decision tree based on Bayesian optimization (BO-FDT) is proposed to identify LOS/NLOS for UWB signals in this article. The main contributions of this article are summarized as follows.

1) Some important features are extracted from the CIRs of UWB signals, and fuzzy c-means (FCM) and triangular membership function are adopted to fuzzify the features. Then, fuzzy features are used as an input to construct FDT for LOS/NLOS identification.
2) Bayesian optimization algorithm is used to select some key parameters in the stage of constructing FDT, and in order to realize Bayesian optimization algorithm, an advanced probability distribution modeling strategy is adopted.
3) Extended experiments to evaluate the relationship between the number of training samples and achieved performance and quantitatively analyze the computational complexity of the proposed method are conducted, and static and dynamic experiments are designed and conducted to evaluate the identification performance of the proposed method, and the identification performance of BO-FDT is compared with other five commonly used methods to verify the advantages of the proposed method.

The remainder of this article is organized as follows. In Section II, the UWB channel model and the extraction of channel statistical features are introduced. In Section III, the construction of FDT and its application to LOS/NLOS identification are presented in detail. The details of the experiments and results analysis are shown in Section IV. In Section V, the conclusions of this article are drawn and future work is given.

II. Channel Model and Feature Extraction

Wireless channel model jointly designed by Saleh and Valenzuela is called the S–V model [30]. The modified model becomes the standard UWB channel model recommended by IEEE. The S–V model is established based on the theory of cluster arrival of multipath components. The arrival rate of cluster and the arrival rate of intracluster multipath components obey Poisson distribution. That is, the CIR of UWB system is expressed as [30]

$$r(t) = \sum_{l=0}^{L-1} \sum_{k=0}^{K_l-1} \beta_{l,k} \exp\left(j\theta_{l,k}\right) \delta\left(t - T_l - \tau_{l,k}\right) \tag{1}$$

where L represents the number of arrival clusters, K_l represents the number of arrival multipath components of the lth cluster, $\beta_{l,k}$ represents the path gain of the kth ray of the lth cluster, and $\theta_{l,k}$ represents the phase of the kth ray of the lth cluster. $\delta(\cdot)$ is the Dirac function, T_l is the arrival time of the lth cluster, and $\tau_{l,k}$ represents the arrival time of the kth ray of the lth cluster. Then, the arrival time of the first cluster is $T_0 = 0$ and the arrival time of the first ray of the lth cluster is $\tau_{l,0} = 0$.

According to the basic operation principle of the UWB device, the received UWB signal waveform is described as

$$h(t) = \int_{-\infty}^{\infty} r(\tau) s(t - \tau) d\tau \tag{2}$$

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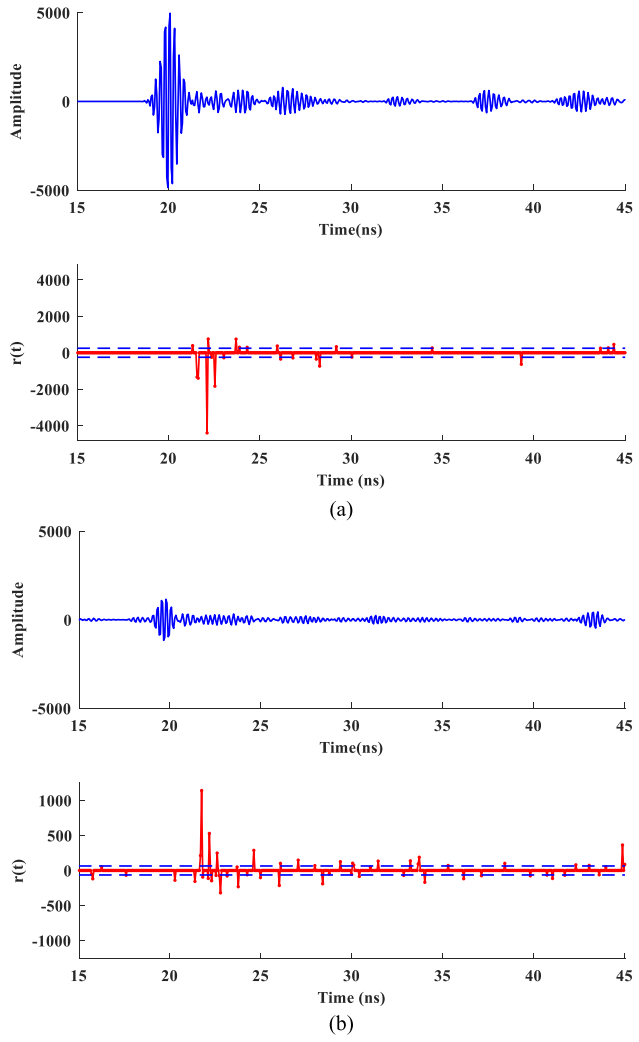


Fig. 1. (a) and (b) Received signal waveforms and the corresponding CIRs under LOS and NLOS conditions, respectively. (The time interval between two adjacent samples is 61 ps, and the threshold is 25 dB.)

where $s(t)$ is the UWB signal waveform transmitted from the antenna.

According to (2), the CIR $r(t)$ can be calculated by deconvolution of received signal waveform $h(t)$ and template signal waveform $s(t)$. Deconvolution can be implemented by the CLEAN algorithm [31]. Notably, the deconvolution process is not applied to every data point of the received signal waveform. Instead, if the received signal strength is less than a threshold, it will be considered too weak to be reliable and thus ignored. The threshold in this article is 25 dB. Fig. 1 shows the received signal waveforms and the corresponding CIRs under LOS and NLOS conditions.

Before constructing the FDT, eight channel features were extracted from the CIRs. These features are total energy (TE), MA, normalized strongest path energy (NSPE), rise time (RT), signal-to-noise ratio (SNR), mean excess delay (MED), root-mean-square (RMS) delay spread, and KUR. The calculation formulas of these features are given in Table I. Combining previous studies and the difference in CIRs under LOS and NLOS conditions, the reasons for selecting these eight features can be summarized as follows.

TABLE I
EXTRACTED FEATURES AND CORRESPONDING FORMULAS

feature	*formula*
TE^a	$\varepsilon = \sum_{i=1}^{N} \lvert r(t_i)\rvert^2$
MA	$r_{max} = max\{\lvert r(t_i)\rvert\}$
$NSPE$	$\gamma = r_{max}^2/\varepsilon$
RT^b	$t_{rise} = t_{upper} - t_{lower}$
SNR^c	$\delta_{snr} = 20\,log_{10}(r_{max}/\sigma_n)$
MED	$\tau_{med} = \frac{1}{\varepsilon}\sum_{i=1}^{N}(t_i\lvert r(t_i)\rvert^2)$
RMS	$\tau_{rms} = \frac{1}{\varepsilon}\sum_{i=1}^{N}[(t_i - \tau_{med})^2\lvert r(t_i)\rvert^2]$
KUR^d	$\kappa = \frac{1}{N\sigma_r^4}\sum_{i=1}^{N}(\lvert r(t_i)\rvert - \mu_r)^4$

a N is the number of sampling points of the CIR $r(t)$. $r(t_i)$ denotes the amplitude of ith sampling point.

b $t_{upper} = min\{t_i: \lvert r(t_i)\rvert \geq 0.9r_{max}\}$ and $t_{lower} = min\{t_i: \lvert r(t_i)\rvert \geq 0.1r_{max}\}$.

c σ_n denotes standard deviation of thermal noise.

d $\mu_r = \frac{1}{N}\sum_{i=1}^{N}\lvert r(t_i)\rvert$ and $\sigma_r^2 = \frac{1}{N}\sum_{i=1}^{N}(\lvert r(t_i)\rvert - \mu_r)^2$.

1) UWB signal propagates under NLOS conditions, with smaller energy and amplitude. In addition, under LOS conditions, the strongest path of signal corresponds to the first path, while under NLOS conditions, weak components tend to precede the strongest path, resulting in longer RT [15], [32].
2) KUR represents the peak state of the signal and reflects the statistical characteristic of the amplitude of the CIR. The CIR with high KUR is more likely to be in LOS conditions. MED and RMS delay spread represent delay information of multipath channels [33].
3) The reflected, diffracted, and scattered signals add up after the first path energy arrives, thus resulting in a lower standard deviation. Due to the difference in energy divergence between LOS and NLOS conditions, the remaining features are selected [34].

The magnitude of different features varies, which affects the comparability of data. Therefore, the min–max method [35] is adopted to normalize data. This normalization method is a linear transformation, which will not cause the loss of original information and change the numerical ranking of original features. Given a feature value, it is normalized by

$$x_{\text{new}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3}$$

where x_{new} is the normalized value, and $x_{\max}$ and $x_{\min}$ are the maximum and minimum of all values of the feature, respectively.

Note that when normalizing the test data, the parameters $x_{\max}$ and $x_{\min}$ are the same as that in the training data.

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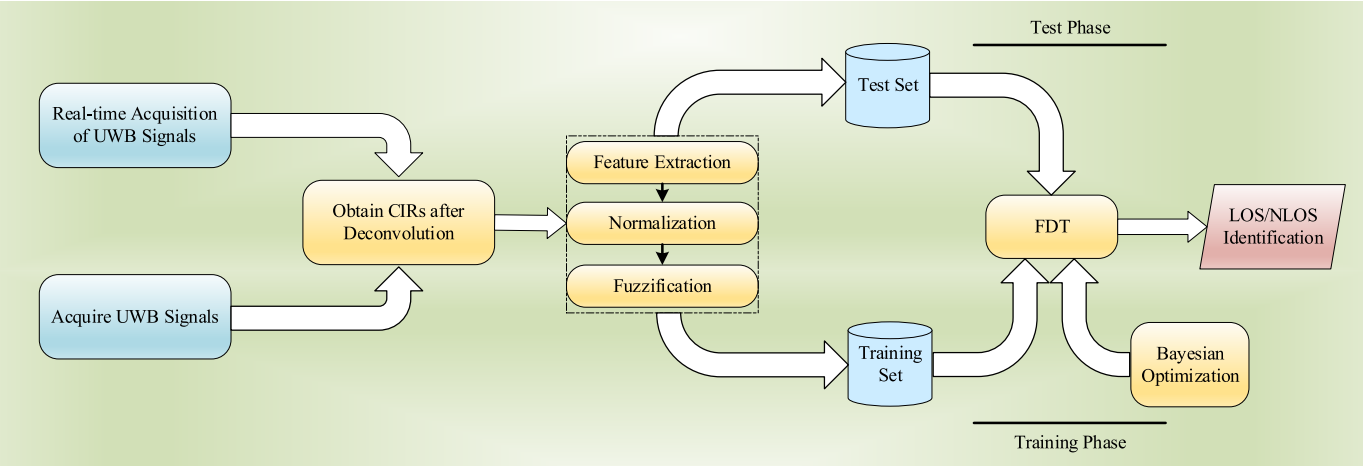


Fig. 2. Overall framework for LOS/NLOS identification.

III. Proposed LOS/NLOS Identification Method

The LOS/NLOS identification performance of decision tree (DT) was evaluated in [24]. FDT is a combination of DT and fuzzy logic to deal with fuzzy and uncertain information. Compared with the traditional DT, FDT has certain advantages in volatile environment. Fig. 2 shows the overall framework of the proposed method in this article. The difference between FDT and DT is mainly in two aspects: 1) the input features are fuzzy features, and each original feature corresponds to several linguistic terms and 2) node splitting criteria are different [36], [37], [38]. This section will detail the construction of BO-FDT.

A. Fuzziication With Fuzzy C-Means and Triangular Membership Function

According to Zadeh's fuzzy set theory [39] and its application [40], the numeric feature X_i agrees with a vector of real numbers $(x_1, x_2, \ldots, x_\kappa)$, where κ is the number of samples. Each element in the vector represents a sample. After fuzzification, X_i is transformed into fuzzy feature $A_i (i = 1, 2, \ldots, n)$, with n being the number of features, and here $n = 8$. Each fuzzy feature A_i consists of $m_i (m_i \geq 2)$ linguistic terms. The jth linguistic term of A_i is represented by a fuzzy set $A_{i,j} (j = 1, 2, \ldots, m_i)$. $A_{i,j}$ is determined by membership function $\mu_{A_{i,j}}(x) : X_i \to \langle 0, 1 \rangle$, which defines membership degree $\mu_{A_{i,j}}(x)$ for each $x \in X_i$. Membership degree provides an estimate of the degree to which element x belongs to $A_{i,j}$. In fact, the fuzzy set $A_{i,j}$ is defined as an ordered set of x and membership degree, which is $A_{i,j} = \{(x, \mu_{A_{i,j}}(x)), x \in X_i\}$.

Transforming numeric feature into corresponding fuzzy feature requires specifying membership function, that is to define the function $\mu_{A_{i,j}}(x)$. The common membership functions are triangular membership function, trapezoidal membership function, and Gaussian membership function [41]. Considering simple calculation and execution, the triangular membership function was selected as the initial choice of fuzzification. The triangular membership function is defined as follows [36]:

$$\mu_{A_{i,j}}(x) = \begin{cases} 1, & x \leq \tilde{C}_{i,1} \\ \dfrac{\tilde{C}_{i,2} - x}{\tilde{C}_{i,2} - \tilde{C}_{i,1}} & \tilde{C}_{i,1} \leq x \leq \tilde{C}_{i,2} \quad (\text{if } j = 1) \\ 0, & x \geq \tilde{C}_{i,2} \end{cases} \tag{4}$$

$$\mu_{A_{i,j}}(x) = \begin{cases} 0, & x \leq \tilde{C}_{i,j-1} \\ \dfrac{x - \tilde{C}_{i,j-1}}{\tilde{C}_{i,j} - \tilde{C}_{i,j-1}}, & \tilde{C}_{i,j-1} \leq x \leq \tilde{C}_{i,j} \\ \dfrac{\tilde{C}_{i,j+1} - x}{\tilde{C}_{i,j+1} - \tilde{C}_{i,j}}, & \tilde{C}_{i,j} \leq x \leq \tilde{C}_{i,j+1} \\ 0, & x \geq \tilde{C}_{i,j+1} \end{cases} \quad (\text{if } 1 < j < m_i) \tag{5}$$

$$\mu_{A_{i,j}}(x) = \begin{cases} 0, & x \leq \tilde{C}_{i,m_i-1} \\ \dfrac{x - \tilde{C}_{i,m_i-1}}{\tilde{C}_{i,m_i} - \tilde{C}_{i,m_i-1}}, & \tilde{C}_{i,m_i-1} \leq x \leq \tilde{C}_{i,m_i} \\ 1, & x \geq \tilde{C}_{i,m_i}. \end{cases} \quad (\text{if } j = m_i) \tag{6}$$

For numerical feature X_i, the number of linguistic terms of the corresponding fuzzy feature A_i is determined by the number of clustering centers of X_i, then a set $\mathbf{C}_i = \{C_{i,1}, C_{i,2}, \ldots, C_{i,m_i}\}$ composed of these clustering centers, whose elements are arranged in ascending order as $\tilde{C}_i = \{\tilde{C}_{i,1}, \tilde{C}_{i,2}, \ldots, \tilde{C}_{i,m_i}\}$. For example, suppose that the number of clustering centers of X_1 is 4, then $m_1 = 4$. Obviously, for each $x \in X_1$, x has four linguistic terms after fuzzification. Conveniently, the corresponding semantics of these four clustering centers can be called as "low," "medium," "high," and "very high"; then, the triangular membership function is shown in Fig. 3.

After the membership function is selected, it is necessary to find the appropriate clustering centers. Usually, the number of clustering centers of different features can be determined empirically. However, since the number of clustering centers has a significant impact on learning efficiency and model performance [42], therefore, this article treats the number of

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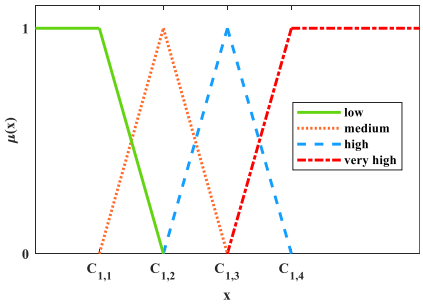


Fig. 3. Triangular membership function.

clustering centers of each feature as an algorithm parameter, which is to be optimized. Meanwhile, FCM is adopted to determine the appropriate clustering centers for each feature [43]. Compared with other clustering algorithms, FCM provides more flexible clustering results. For any of the eight features in this article, they can hardly be divided into clearly separated clusters according to the number of clustering centers. Then, FCM can be used to assign a certain weight to an object and a cluster, indicating the degree to which the object belongs to the cluster. Therefore, FCM algorithm with natural, nonprobability characteristics is adopted. The algorithm searches clustering centers of each feature by minimizing the objective function, and the process can be expressed as

$$\mathbf{C}_i = \underset{\mathbf{C}_i}{\operatorname{argmin}} \sum_{u=1}^{m_i} \sum_{v=1}^{\kappa} \mu_{uv}^2 \left\| x_{i,v} - C_{i,u} \right\|^2 \tag{7}$$

where m_i and κ are defined as above, $x_{i,v}$ represents the vth value of X_i, $C_{i,u}$ denotes the uth clustering center of X_i, μ_{uv} represents the degree to which $x_{i,v}$ belongs to a cluster centered at $C_{i,u}$ and satisfies $\sum_{u=1}^{m_i} \mu_{uv} = 1$, $v = 1, 2, \ldots, \kappa$. $\|\cdot\|$ represents the distance measure, which uses Euclidean distance in this article.

The degree μ_{uv} and clustering center $C_{i,u}$ are iteratively updated by

$$\mu_{uv} = 1 \Big/ \sum_{k=1}^{m_i} \left(\frac{\left\| x_{i,v} - C_{i,u} \right\|}{\left\| x_{i,v} - C_{i,k} \right\|} \right)^2 \tag{8}$$

$$C_{i,u} = \sum_{v=1}^{\kappa} \mu_{uv}^2 x_{i,v} \Big/ \sum_{t=1}^{\kappa} \mu_{ut}^2. \tag{9}$$

The algorithm stops iterating if one of the following conditions is met: 1) the total number of iterations reaches the maximum and 2) the difference of degrees between two adjacent iterations is less than a predefined threshold. Finally, according to the above steps, fuzzy datasets of {TE, MA, NSPE, RT, SNR, MED, RMS, KUR} can be obtained.

B. Fuzzy Decision Tree

In this article, the induction of FDT is based on the classification uncertainty. The classification uncertainty can directly measure the quality of classification rules at the decision node, and the node splitting is based on reducing the classification uncertainty with accumulated fuzzy evidences [36].

The cardinality of a fuzzy set $A_{i,j}$ is defined as, $M(A_{i,j}) = \sum_{x \in \mathbf{X}_i} \mu_{A_{i,j}}(x)$. Assuming that a nonleaf node B is extended, the candidate fuzzy features are $A_i (i = 1, 2, \ldots, n)$ and linguistic terms are $A_{i,j} (j = 1, 2, \ldots, m_i)$. For the target attribute A_{n+1}, $A_{n+1,1}$ represents LOS condition and $A_{n+1,2}$ denotes NLOS condition; then, the target attribute of each sample can be expressed as: LOS: [1, 0] or NLOS: [0, 1]. For each $A_{i,j}$, its relative frequency on the nonleaf node B with respect to the class $A_{n+1,l} (l = 1 \text{ and } 2)$ is defined as

$$\begin{aligned} p_{i,j}^l &= \frac{M\left(A_{i,j} \cap A_{n+1,l} \cap B\right)}{M\left(A_{n+1,l} \cap B\right)} \\ &= \frac{\sum_{x \in X_i} \min\left(\mu_{A_{i,j}}(x), \mu_{A_{n+1,l}}(x), \mu_B(x)\right)}{\sum_{x \in X_i} \min\left(\mu_{A_{n+1,l}}(x), \mu_B(x)\right)}. \end{aligned} \tag{10}$$

The relative frequencies are normalized to get $\tau_{i,j}^l = p_{i,j}^l / \max_{l=1,2} p_{i,j}^l$; then, $\tau_{i,j}^l$ are arranged in ascending order as $\pi_{i,j}^l$. Then, for each $A_{i,j}$, its classification uncertainty on the nonleaf B is

$$G\left(A_{i,j}\right) = \sum_{l=1}^{2} \left(\pi_{i,j}^l - \pi_{i,j}^{l+1} \right) \ln l \tag{11}$$

where $\pi_{i,j}^3 = 0$.

Therefore, for each fuzzy feature A_i, its average classification uncertainty on nonleaf nodes B is

$$G(A_i) = \sum_{j=1}^{m_i} \omega_j G\left(A_{i,j}\right) \tag{12}$$

$$\omega_j = \frac{M\left(A_{i,j} \cap B\right)}{\sum_{t=1}^{m_i} M\left(A_{i,t} \cap B\right)}. \tag{13}$$

In the process of constructing FDT, two key parameters are used to control the splitting of nodes, namely, the significant level α and the truth level threshold β. As a pruning parameter, the parameter $\alpha (0 \le \alpha \le 1)$ is used to eliminate unimportant branches and leaves. For a fuzzy set $\tilde{A}$, its membership function is $\mu_{\tilde{A}}(x)$; then, under the significance level α, the fuzzy set $\tilde{A}_\alpha$ is defined as

$$\tilde{A}_\alpha = \begin{cases} \mu_{\tilde{A}}(x), & \mu_{\tilde{A}}(x) \ge \alpha \\ 0, & \mu_{\tilde{A}}(x) < \alpha \end{cases}. \tag{14}$$

β controls the growth of the tree. If the classification truth level of the current node is greater than the threshold β, the node becomes a leaf node; otherwise, the node continues to split. Usually, the threshold β satisfies $0.5 < \beta < 1$. The classification truth level of a node refers to the maximum value of the classification truth level of each category. For node B, the truth level of category $A_{n+1,l}$ is

$$\hat{\beta} = \frac{M\left(B \cap A_{n+1,l}\right)}{M(B)}. \tag{15}$$

Usually, there are no definite rules for the selection of parameters α and β, so they are selected empirically. In order to improve the performance of FDT and considering the practical application, the Bayesian optimization algorithm is used to find the appropriate parameters.

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C. Bayesian Optimization

In order to find the appropriate values of the number of clustering centers of each feature, α and β, Bayesian optimization algorithm is adopted to determine these parameters. Bayesian optimization algorithm establishes probability distribution model based on the historical observation information of objective function, and then, changes in objective function can be obtained. Historical observation information refers to the historical information before a certain moment in the parameter optimization process. Historical observation information as prior knowledge, which is used to approximate the posterior distribution of the objective function to adjust the parameters. The main reasons for using Bayesian optimization algorithm are that: first, the optimization parameters include continuous and discrete values. Second, the optimization algorithm is based on randomness and probability distribution, which is not easy to fall into local optimum. In addition, the algorithm makes full use of prior knowledge and historical observation information to accelerate the optimization process. In order to realize Bayesian optimization algorithm, the probability distribution modeling strategy based on tree-structured Parzen estimator (TPE) is used [44].

In the process of Bayesian optimization, the algorithm will determine the search direction of the next iteration according to the position of the best parameters for the current search, and expected improvement (EI) is often used as the acquisition function to judge. The expression of EI is given by

$$\mathrm{EI}_{\rho}(\mathbf{x}) = \int_0^1 \max(\rho - y, 0)\, p(y \mid \mathbf{x}) dy \tag{16}$$

where y denotes the classification error, and $\mathbf{x}$ is the parameter vector. For the LOS/NLOS identification method based on FDT in this article, $\mathbf{x}$ consists of the numbers of clustering centers corresponding to eight features, α and β. ρ is the given classification error threshold, and $p(y|\mathbf{x})$ is the conditional PDF.

In fact, EI denotes the expectation, that is, how much the classification performance improves relative to the threshold when the parameter vector is given. Therefore, minimizing the classification error is equivalent to maximizing EI, and the parameter vector is iteratively updated by

$$\mathbf{x}_{\text{new}} = \underset{\mathbf{x}}{\operatorname{argmax}}\, \mathrm{EI}_{\rho}(\mathbf{x}). \tag{17}$$

In the process of using the TPE strategy to establish the probability distribution model, the conditional PDF $p(\mathbf{x}|y)$ is divided into two individual PDF

$$p(\mathbf{x} \mid y) = \begin{cases} h(\mathbf{x}), & y < \rho \\ g(\mathbf{x}), & y \geq \rho. \end{cases} \tag{18}$$

That is, for the parameter vectors on both sides of the threshold ρ, different distributions are constructed, respectively. $h(\mathbf{x})$ represents the PDF of the parameter vectors, whose classification error y is less than ρ in all the parameter vectors that have been searched, and $g(\mathbf{x})$ represents the PDF of the remaining parameter vectors.

After this division, then constructing

$$p(\mathbf{x}) = \int_{\Re} p(\mathbf{x} \mid y) p(y) dy = \lambda h(\mathbf{x}) + (1 - \lambda) g(\mathbf{x}) \tag{19}$$

where $\lambda = p(y < \rho)$, and $\Re$ is the historical observation information, that is, the totality of classification errors.

According to the Bayesian conditional probability formula, (16) can be transformed into

$$\mathrm{EI}_{\rho}(\mathbf{x}) = \int_0^1 \max(\rho - y, 0) \frac{p(\mathbf{x} \mid y) p(y)}{p(\mathbf{x})} dy. \tag{20}$$

Substituting (18) and (19) into (20) and then performing some mathematical operations

$$\mathrm{EI}_{\rho}(\mathbf{x}) = \frac{\lambda \rho h(\mathbf{x}) - h(\mathbf{x}) \int_0^{\rho} y p(y) dy}{\lambda h(\mathbf{x}) + (1 - \lambda) g(\mathbf{x})} = \frac{\lambda \rho - \int_0^{\rho} y p(y) dy}{\lambda + (1 - \lambda) g(\mathbf{x}) / h(\mathbf{x})}. \tag{21}$$

Therefore, the Bayesian optimization algorithm based on TPE finds the parameter vector corresponding to the minimum classification error, which is the parameter vector that minimizes $g(\mathbf{x})/h(\mathbf{x})$.

IV. Experimental Results and Analysis

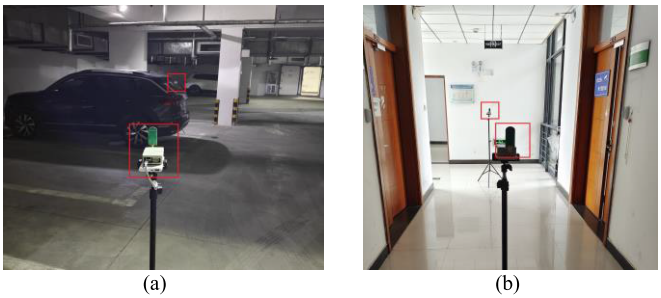
In this section, we will evaluate the performance of the proposed LOS/NLOS identification method in harsh and volatile environments. Two experiments were conducted in this article, one is a static experiment and the other is a dynamic experiment. Different from previous studies, we collected training data and test data independently in each experiment, instead of uniformly collecting and then dividing the data. In addition, meanwhile, the existing five algorithms, namely, DT [19], least-squares support-vector machines (LS-SVM) [15], [45], RF [19], CNN [24], and MWT-CNN [24], are selected for performance comparison with the proposed method. In addition, extended experiments were conducted to analyze the relationship between the number of training samples and achieved performance and evaluate the computational complexity of the proposed method.

A. Experimental Setup and Data Collection

1) Experimental Apparatus: In these two experiments, PulsOn 440 modules were used, which are the UWB wireless transceivers manufactured by Time Domain Company. The operating band is from 3.1 to 4.8 GHz, which can realize communication between two or more modules and the transmission power of the module is very low. During experiments, each received UWB signal waveform was collected with a time window of 100 ns, that is, the waveform was recorded from 20 ns before the leading edge to 80 ns after the leading edge. The sampling rate is 61 ps. The USB is used to connect module and computer to control signal transmission, reception, and data recording. The softwares used for executing algorithms are MATLAB2021b and python 3.10.1. The computer used is equipped with Windows 10 (x64), i5-8250U (1.8 GHz), and RAM 8.00 GB.

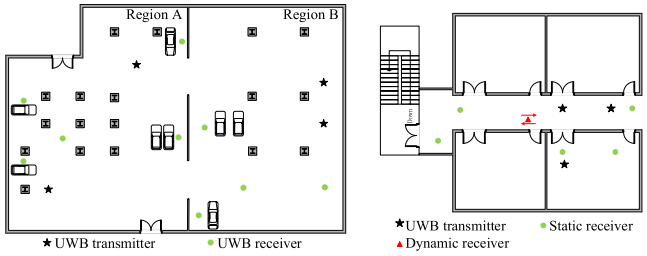
2) Static Experiment: The static experimental field is an underground parking lot of the China University of Mining and Technology (CUMT). The experimental scenario is shown in Fig. 4(a), and the layout of this scenario is shown in Fig. 5(a). According to the internal structure of the underground parking lot, the experimental region is divided into regions A and B [as shown in Fig. 5(a)]. Regions A and B have similar structures and have the same types of objects that cause

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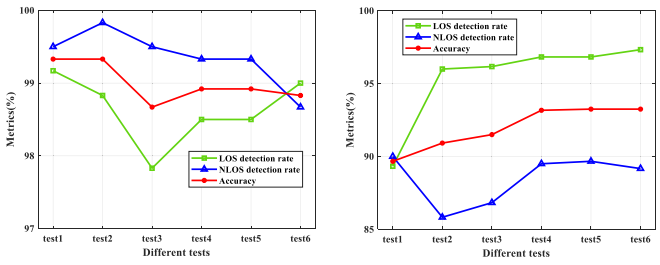
(a) (b)

Fig. 4. Actual experimental fields. (a) Underground parking lot (static experiment). (b) Office building (dynamic experiment).



(a) (b)

Fig. 5. Layout of experimental fields. (a) Layout of static experimental field. (b) Layout of dynamic experimental field. The static receiver indicates that the receiver is in a static state, and the dynamic receiver indicates that the experimental personnel carries the receiver back and forth.



(a) (b)

Fig. 6. Comparison of LOS/NLOS identification performance for different tests in (a) static and (b) dynamic extended experiments.

NLOS propagation of the signal, such as cars and pillars. This experimental design is to verify the adaptability of the proposed method to similar environments and its robustness to small changes in the environment. The training data were collected in region A. Under NLOS conditions, the obstacles are cars and pillars. Then, the testing data were collected in region B.

3) Dynamic Experiment: The dynamic experiment was conducted in an office building of CUMT. The experimental scenario is shown in Fig. 4(b), and the layout of this scenario is shown in Fig. 5(b). In order to deal with and model the dynamic changes and uncertainties that may occur in the indoor environment in reality, first of all, the training data containing LOS and different NLOS propagation channels were collected. The types of obstacles are corners, wooden doors, walls, and human bodies. Then, the testing data were collected in a dynamic way. As shown in Fig. 5(b), the experimental personnel carried the UWB receiver and moved back and forth along a straight line in the corridor.

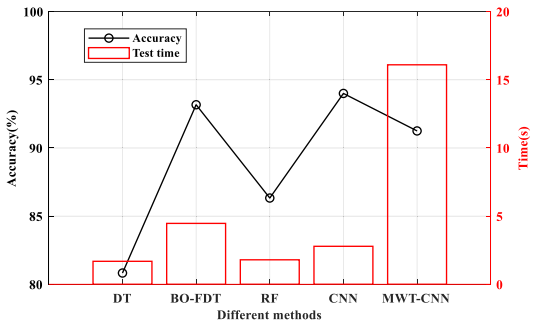


Fig. 7. Comparison of test time and accuracy of LOS/NLOS identification with different methods.

B. Classification Performance and Complexity Analysis

Similar to [21], the metrics of classification performance used in this article are LOS detection rate, NLOS detection rate, correct rate, and correct rate is treated as accuracy in this article. In order to analyze the influence of the number of samples contained in the training set on the accuracy of the classification model proposed in this article, for both static and dynamic experiments, we set up six comparative tests. In the static experiment, the number of samples in the training set of six different tests is 100, 300, 600, 1600, 2600, and 3600, respectively. Conveniently, tests 1–6 represent these six tests, respectively. The comparison results are shown in Fig. 6(a). The classification accuracy of the six tests is 99.33%, 99.33%, 98.67%, 98.92%, 98.92%, and 98.83%, respectively. It shows that the accuracy changes little with the number of samples contained in the training set, which is less than 1%. Note that as the number of samples decreases, the accuracy increases slightly. We consider that this performance change may be related to the experimental environment. In this static experiment, satisfactory performance can be achieved when the number of samples contained in the training set is more than 100. In addition, in the dynamic experiment, the number of samples in the training set of six different tests is 300, 1600, and 2600 and 3600, 4600, and 5600, respectively. As shown in Fig. 6(b), the classification accuracy of the six tests is 89.67%, 90.92%, 91.5%, 93.17%, 93.25%, and 93.25%, respectively. Clearly, increasing the number of training samples can only improve classification accuracy rather marginally when the number of training samples is greater than 3600. When the number of samples is less than 1600, the accuracy is reduced significantly. In this dynamic experiment, it is more appropriate that the number of samples contained in training set is higher than 1600. In addition, without loss of generality, in the subsequent experiments, 3600 samples were adopted to support the training of the model in both static and dynamic experiments.

In addition, as shown in Fig. 7, the time consumption and accuracy of different methods were compared in the test phase for the dynamic experiment. The identification of LOS/NLOS using the proposed algorithms, such as DT, RF, CNN, and morlet wavelet transform-CNN (MWT-CNN), was performed on the python 3.10.1 platform. Results show that the accuracy of DT and RF in the test phase is much lower than that of the other three methods. The accuracy of the proposed method, CNN and MWT-CNN, is 93.17%,

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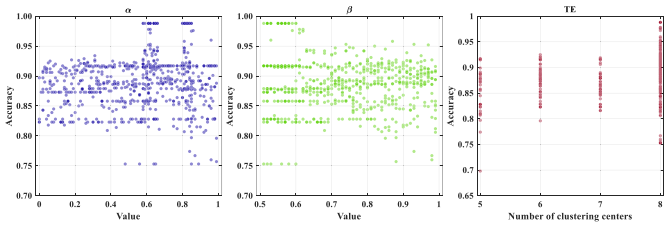


Fig. 8. Dependency between accuracy and different parameters for static experiment.

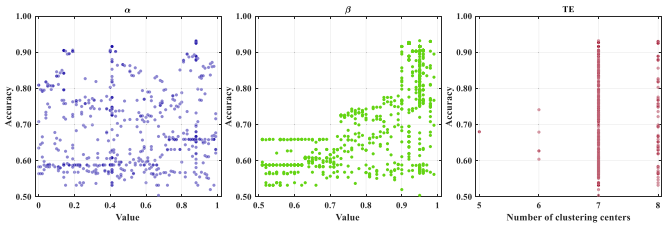


Fig. 9. Dependency between accuracy and different parameters for dynamic experiment.

94.00%, and 91.25%, respectively. In addition, the calculation time of the proposed method is 4.47 s, which is less than the calculation time of MWT-CNN but higher than the other three methods. The main reason is that the proposed method needs to convert the original features into fuzzy features during execution. However, for the proposed method, the time required to identify each sample is only at the millisecond level, which is sufficient to meet the time requirements of real-time positioning.

C. Effect of Parameters on Identification Performance

In the process of constructing FDT, α, β, and the number of clustering centers corresponding to each feature have an impact on the size and performance of FDT. Bayesian optimization algorithm is thus used to find the appropriate parameters to obtain the optimal size of the FDT and LOS/NLOS identification performance. Figs. 8 and 9 show the effect of the parameters in the Bayesian optimization process on the final test accuracy in two experiments. For simplicity, only the relationship between the number of clustering centers of feature **TE** and accuracy is shown. Fig. 8 shows the dependency between the parameters and the accuracy in static experiment. In the iterative process, when the parameter α is between 0.50 and 0.85, and the parameter β is between 0.50 and 0.60, and the identification accuracy of the method is higher than 95%. For the feature **TE**, when the number of clustering centers is 8, the accuracy is greater than 95%. Finally, in the static experiment, the optimal identification performance is achieved when $\alpha = 0.59$ and $\beta = 0.60$, and the numbers of clustering centers for these eight features are 8, 7, 8, 7, 8, 8, 8, and 8, respectively. In addition, Fig. 9 shows that when α is around 0.88, β is around 0.95, and the number of clustering centers of the feature **TE** is 7, the accuracy is higher than 92%. Therefore, in dynamic experiment, optimal performance is obtained when $\alpha = 0.88$, $\beta = 0.95$, and the numbers of clustering centers corresponding to the eight features are 7, 5, 5, 5, 8, 6, 6, and 7, respectively.

D. Performance Comparison Among Different Algorithms

To further evaluate the performance of the proposed BO-FDT in the identification of LOS/NLOS propagation for UWB signals, five existing algorithms were evaluated against the proposed method. In [19], it is shown that the performance of the classification algorithm will improve as the number of features increases, but for SVM, when the number of features reaches a certain number, its performance will decrease as the number of features increases. This is because as the number of features increases, the likelihood of high correlation increases significantly. Therefore, for LS-SVM, the feature set that performs the best was chosen in these two experiments. For the static experiment, the feature set with the best performance is {RT, SNR, MED, RMS}; for the dynamic experiment, the feature set with the best performance is {TE, MA, NSPE, RT, SNR, MED}. Now, consider DT and RF, in general, redundant features do not adversely affect the performance of these two methods because DT is selective for features. For example, among two redundant features, if one of them has been selected as the splitting feature, the other will be ignored [46]. Similar to DT, this strategy can also be taken for RF. Therefore, DT and RF will use all the eight features for classification. Combined with the above analysis of DT and experimental observation, all features are retained during the execution of BO-FDT for LOS/NLOS identification, which avoids the problem that some other methods (such as SVM and LS-SVM) require feature selection.

Based on these eight features, two pruned FDTs were established in different experiments. Fig. 10 shows the FDT for the static experiment, while Fig. 11 shows the FDT for the dynamic experiment. Fig. 10 shows that the depth of the FDT is 1, and the root node takes the feature **TE** as the splitting feature and splits according to its linguistic terms. Fig. 11 shows a more complex FDT with a depth of 3 and the feature **MA** is splitting feature of the root node. In fact, the dynamic experimental field has a more complex environment than the static experimental field. A number of factors, such as human disturbance or wireless signal interference, will inevitably increase the uncertainty of received UWB signals. While the static experimental field is an underground parking lot, its environmental conditions are relatively stable, with less interference from other factors. Therefore, the structure of the FDT at the static experimental field is relatively simple.

Tables II and III show the LOS detection rate, NLOS detection rate, and accuracy of the six algorithms at the two experimental fields, respectively. Table II shows that the NLOS detection rate of LS-SVM and RF reaches 100%. Although the NLOS detection rate of the proposed method is slightly lower than other methods, the LOS detection rate reaches 99%, which is significantly higher than other methods. Also, the overall identification accuracy of BO-FDT is improved by 4.41%, 3.91%, 3.16%, 6%, and 7.50% compared to LS-SVM, DT, RF, CNN, and MWT-CNN. In addition, Table III shows that except for CNN, the LOS and NLOS detection rates and overall identification accuracy of the BO-FDT method

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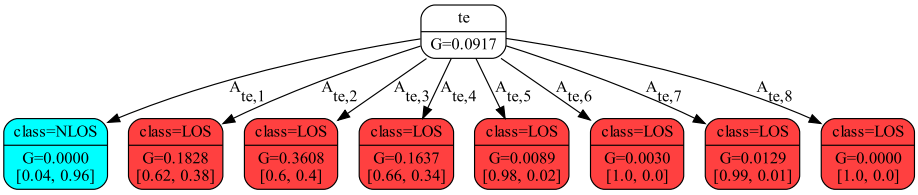


Fig. 10. FDT for LOS/NLOS identification in static experiment. $\alpha = 0.59$ and $\beta = 0.60$. $A_{i,j}$ represents the jth linguistic term of feature i, where $i \in \{\text{TE, MA, NSPE, RT, SNR, MED, RMS, KUR}\}$, $j \in C_i$, and the number of clustering centers corresponding to this feature is C_i.

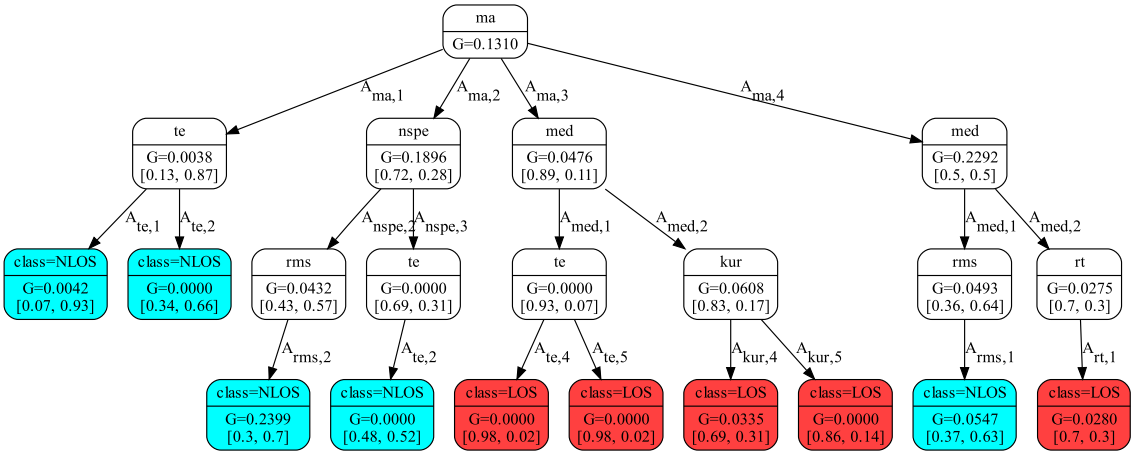


Fig. 11. FDT for LOS/NLOS identification in dynamic experiment. $\alpha = 0.88$ and $\beta = 0.95$.

TABLE II
COMPARISON OF PERFORMANCE OF LOS/NLOS IDENTIFICATION IN STATIC EXPERIMENT

Methods	LOS detection rate (%)	NLOS detection rate (%)	Accuracy (%)
BO-FDT	99.00	98.67	98.83
LS-SVM	88.83	100	94.42
DT	90.50	99.33	94.92
RF	91.34	100.00	95.67
CNN	86.00	99.67	92.83
MWT-CNN	82.83	99.83	91.33

* LS-SVM was performed using the LS-SVMlab toolbox. For LS-SVM, the kernel function is "RBF", gam=9.3943, sig2=0.2885.

TABLE III
COMPARISON OF PERFORMANCE OF LOS/NLOS IDENTIFICATION IN DYNAMIC EXPERIMENT

Methods	LOS detection rate (%)	NLOS detection rate (%)	Accuracy (%)
BO-FDT	96.83	89.50	93.17
LS-SVM	94.00	80.00	87.00
DT	78.50	83.17	80.83
RF	86.50	86.17	86.33
CNN	98.50	89.50	94.00
MWT-CNN	97.00	85.50	91.25

* For LS-SVM, the kernel function is "RBF", gam=30.5386, sig2=0.1082.

are greater than the other methods. The overall identification accuracy of BO-FDT reaches 93.17%, which is 6.17%, 12.34%, 6.84%, and 1.82% higher than that of LS-SVM, DT, RF, and MWT-CNN, respectively. Combined with the results in Tables II and III, the LOS detection rate of the BO-FDT method reaches 95% and the accuracy is higher than 90% in both static and dynamic experiments. In terms of overall performance, except for slightly lower than the CNN method in dynamic experiment, the proposed method has higher LOS/NLOS identification performance in both experiments than the other methods. As mentioned above, the environment of dynamic experiment is more complex than that of static experiment, and it faces more random factors in the online testing phase. Compared with LS-SVM, DT, and RF, the proposed BO-FDT method improves the accuracy more significantly in dynamic experiment than in static experiment, so the advantages of BO-FDT method applied to LOS/NLOS identification under harsh and volatile indoor conditions will be more significant.

V. CONCLUSION

In this article, a method named BO-FDT is proposed for UWB LOS/NLOS identification. In order to simplify the process of parameters selection and further improve the performance of FDT, Bayesian optimization method was adopted to find appropriate parameters. Two conclusions can be drawn from the extended experimental results. First, in our static and dynamic experiments, for the BO-FDT method, in order to obtain a stable performance with accuracy higher than 90%, the number of samples in the training set should be higher than 100 and 3600, respectively. Second, the calculation time of identifying each sample is at the millisecond level, and the real-time positioning system can afford the time consumption. In addition, static and dynamic experiments were conducted.

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By constructing FDTs for LOS/NLOS identification, the identification accuracy of the proposed BO-FDT method in the two experiments is 98.83% and 93.17%, respectively, which is better than 90%. Compared with other methods, BO-FDT has certain advantages, and LOS detection rate in both experiments is better than 95%. These show that the proposed method has stable performance in different scenarios and has certain robustness. The future work may include two aspects. First, other splitting criteria will be considered to construct FDT and their effect on the structure of FDT and identification performance also needs to be analyzed. Second, the problem of how to apply the identification results to UWB localization in indoor environments needs to be solved.

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